

Hardware Component	Deployment Role	Configuration Setting	Model or Backend	Device Assignment	Optimisation Flags	Operational Status (Inferred)	Source
Tesla P40 GPU (Dual)	Local inference server / host	runtime.hf_llm_device_map, runtime.speech_device	Qwen-14B (HF), Whisper (ASR)	auto, cuda:0	4-bit loading (runtime.hf_llm_load_in_4bit), persistence mode ON, GGUF quantization (Q4/Q5/Q8)	Ready and Operational; subject to VAD thresholds	1-3
Xeon 16-core CPU (Dual)	Remote bridge server / local host	runtime.caption_device, runtime.tts_device	BLIP (captioning), pyttsx3/SpeechT5 (TTS), LVTC backend	cpu	NUMA-pinning (numactl)	Operational with high latency on CPU-heavy tasks	1-3
Realtek Microphone Array	Audio sensor input	microphone.device	Distil-Whisper	Windows WASAPI	VAD enabled, Software Preamp (8.0x), AGC	Operational (Threshold tuning required)	3
Peripheral Camera (index 0)	Visual sensor input	--camera-index	BLIP / Captioning	/dev/video0	Saliency-driven triggering	Operational via remote bridge	3, 4
Windows Laptop (Client)	Remote sensor provider	bridge client --server-url	Remote AV Bridge	Not in source	Not in source	Operational via LAN/Tailscale	3, 4
1.92TB weights drive (SSD)	Durable state / cache storage	runtime.offload_dir, HF_HOME	Model Weights Storage	/mnt/weights	ext4/xfs, noatime mount flag	Operational (Permissions/Space fixed)	1-3